> **Real-paper format (unofficial reconstruction).** Section A: 4 MCQs, choose one answer each (3 marks each). Section B: 3 structured questions; show all working on paper and type the key final numbers into the boxes. **40 marks, 45 minutes**, closed book, calculator allowed. Nothing is marked or shown until you press *Finish & mark* or the timer runs out. Then the MCQs and boxes mark themselves, the worked solutions open, and you tick the rubric against your written working.
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> Why this shape: Wen confirmed MCQs plus written questions for Quiz 3 (Week 12 lecture), his Quiz 1 this term was one calculation plus four MCQs, and three seniors recall three calculation questions (Jury K-range, s→z mapping, direct design). See quiz-03-intel.md. Numbers are new and verified by `quiz-03-mock-cd-check.py`. Answers: 3–4 significant figures or an exact fraction.

▼ **Transform pairs provided** (as on the real paper)

$X(s)$	$X(z)$
$\frac{1}{s}$	$\frac{z}{z-1}$
$\frac{1}{s^2}$	$\frac{Tz}{(z-1)^2}$
$\frac{1}{s+a}$	$\frac{z}{z-e^{-aT}}$
$\frac{a}{s(s+a)}$	$\frac{(1-e^{-aT})z}{(z-1)(z-e^{-aT})}$

The ZOH $\frac{1-e^{-Ts}}{s}$ and the z-transform properties are *not* provided.

Section A · Multiple choice

12 marks

A1. A unity-feedback loop has open-loop pulse transfer function $G(z) = \dfrac{0.5(z+0.5)}{(z-0.5)(z-0.2)}$, and the closed loop is stable. The steady-state error to a unit step is

- (A) 0.533
- (B) 0
- (C) 0.348
- (D) 0.652

Answer: C

The loop is type 0, so the step error is finite: $K_p = \lim_{z\to 1} G(z) = \frac{0.5(1.5)}{(0.5)(0.8)} = 1.875$ and $e_{ss} = \frac{1}{1+K_p} = \frac{1}{2.875} = 0.348$ pp. 186–188. (A) is $1/K_p$, (B) would need a pole at $z = 1$, (D) is $K_p/(1 + K_p)$, the final output value. (Closed-loop poles $0.1 \pm 0.583j$, inside the circle ✓.)

A2. $G(s) = \dfrac{5(s+3)}{(s+1)(s+2)(s+6)}$ is converted by pole-zero matching into a *strictly proper* $G(z)$. How many factors $(z + 1)$ appear in the numerator?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Answer: B

$G(s)$ has $n - m = 3 - 1 = 2$ zeros at $s = \infty$. These map to $z = -1$, but a strictly proper $G(z)$ keeps one sample of delay, so it takes $n - m - 1 = 1$ factor $(z + 1)$. A proper result would take $n - m = 2$ pp. 14–15.

A3. Why is $G(z)$ transformed to the w-plane, $w = \frac{2}{T}\frac{z-1}{z+1}$, before a Bode-based phase-lead design?

- (A) To remove the effect of the zero-order hold from the loop

○ (B) Because the w-plane removes the frequency warping of the bilinear rule

• (C) To make an unstable $G(z)$ stable before the design starts

○ (D) Because $G(e^{j\omega T})$ is not rational in ω, while $G(w)$ is, so ordinary Bode asymptote rules apply

Answer: D

On the unit circle $z = e^{j\omega T}$, so G is a function of $e^{j\omega T}$, not a rational function of ω, and straight-line Bode rules fail. The w-transform maps the unit disc onto the left half w-plane, and $G(w)$ is rational in the fictitious frequency v pp. 23–24. (B) is backwards: the w-plane has the same warping, $v = \frac{2}{T}\tan\frac{\omega T}{2}$. The ZOH stays inside $G(z)$, and the transform does not change stability.

A4. In a one-degree-of-freedom loop, the reference response $T(z)$ and the disturbance response $S(z)$ satisfy $S = 1 - T$. What does the two-degree-of-freedom structure add?

• (A) A separate block acting on r, so the reference response can be shaped independently of the feedback part that sets stability and disturbance rejection

○ (B) The ability to cancel plant zeros outside the unit circle safely

○ (C) A guarantee of zero steady-state error without any integrator

○ (D) Fewer delay elements than standard programming

Answer: A

With one block on the error $r - y$, choosing T fixes $S = 1 - T$: there is one free function, not two. The 2-DOF form $U = \gamma\frac{T_c}{R_c}R - \frac{S_c}{R_c}Y$ lets R_c, S_c place the closed-loop poles (stability, disturbance) while T_c, γ shape the reaction to r pp. 84–87. Unstable plant zeros still cannot be cancelled in any structure.

Section B · Structured questions

28 marks

EE6203 Quiz 3 · Mock D **Marked 6/40** Reset

B1. (10 marks) A digital gain K controls a process in a unity-feedback loop. Including one sample of computation delay, the open-loop pulse transfer function is

$$G(z) = \frac{K}{z(z-1)(z-0.5)}, \qquad T = 0.1\ \text{s}.$$

(a) Use the Jury test to find the range of K for closed-loop stability. [7]

(b) With $K = 0.2$, find K_v and the steady-state error to a unit ramp. [3]

Final answers:

upper limit of K = 1 × key 0.2808 K_v = 4 ✓

e_{ss} = 0.25 ✓

Worked solution

(a) $1 + G = 0$ gives $P(z) = z^3 - 1.5z^2 + 0.5z + K$, so $a_0 = 1,\ a_1 = -1.5,\ a_2 = 0.5,\ a_3 = K$. This is third order, so the table row is needed as well pp. 164–166:

- (1) $|a_3| < a_0$: $-1 < K < 1$.
- (2) $P(1) = 1 - 1.5 + 0.5 + K = K > 0$.
- (3) n odd, so $P(-1) < 0$: $-1 - 1.5 - 0.5 + K = K - 3 < 0 \rightarrow K < 3$.
- (4) Row 3 uses $b_k = a_n a_{k+1} - a_0 a_{n-1-k}$: $b_0 = a_3 a_1 - a_0 a_2 = -1.5K - 0.5$ and $b_2 = a_3^2 - a_0^2 = K^2 - 1$. The condition is $|b_2| > |b_0|$. With $0 < K < 1$ this reads $1 - K^2 > 1.5K + 0.5$, i.e. $K^2 + 1.5K - 0.5 < 0$, so $K < \frac{-1.5+\sqrt{4.25}}{2} = 0.2808$.

Intersection: $0 < K < 0.2808$. Here the table row is the binding condition. Stopping after (1)–(3) gives the wrong answer $0 < K < 1$. Spot check at $K = 0.2$: $|b_2| = 0.96 > |b_0| = 0.8$ ✓.

(b) $K_v = \frac{1}{T}\lim_{z\to 1}\frac{z-1}{z}\cdot\frac{K}{z(z-1)(z-0.5)} = \frac{1}{0.1}\cdot\frac{K}{0.5} = 20K$ pp. 186–187. At $K = 0.2$ (stable ✓): $K_v = 4$, $e_{ss} = 0.25$.

> Common slips: using $P(-1) > 0$ (the even-n rule); stopping at $K < 1$ without building the b row; mixing up the b_k index order (the z^0 column of row 3 is $b_{n-1} = b_2$).

Mark yourself (tick what your working shows):

- ☐ (a) characteristic equation

- ☐ (a) conditions (1)–(3), including the odd-n sign (1 each)
- ☐ (a) b_0 and b_2 correct
- ☐ (a) $|b_2| > |b_0|$ solved to $K < 0.281$
- ☐ (a) final range $0 < K < 0.281$
- ☐ (b) $K_v = 4$ including $1/T$
- ☐ (b) $e_{ss} = 0.25$

B2. (8 marks) An analog lead controller $D(s) = \dfrac{4(s+2)}{s+8}$ is to be implemented with $T = 0.1$ s.

(a) Find $D(z)$ by the bilinear transformation in the form $K_d \dfrac{z - z_0}{z - p_0}$. [3]

(b) Find the pole-zero matched $D(z) = \alpha \dfrac{z - z_1}{z - p_1}$ with the DC gain matched. Give α. [2]

(c) Write the direct-programming difference equation for your answer to (a), with input $e(k)$ and output $u(k)$. If $e(k) = 1$ for $k \geq 0$ (and 0 before), find $u(0)$ and $u(1)$. [3]

Final answers:

$K_d =$ 1 × key 3.143 $z_0 =$ 18 × key 0.8182

$p_0 =$ 3 × key 0.4286 $\alpha =$ 0.55 × key 3.038

$u(0) =$ -7 × key 3.143 $u(1) =$ 3 × key 1.918

Worked solution

(a) $\frac{2}{T} = 20$, $s = 20\frac{z-1}{z+1}$ pp. 7–8. Then $s + 2 = \frac{22z-18}{z+1}$ and $s + 8 = \frac{28z-12}{z+1}$, and the $(z+1)$ cancels:

$$D(z) = \frac{4(22z-18)}{28z-12} = \frac{22z-18}{7z-3} = \frac{22}{7} \cdot \frac{z - 0.8182}{z - 0.4286}.$$

$K_d = 3.143$, $z_0 = 0.8182$, $p_0 = 0.4286$. DC check: $D(1) = \frac{4}{4} = 1 = D(s{=}0) = \frac{4 \cdot 2}{8}$ ✓.

(b) $z_1 = e^{-0.2} = 0.8187$, $p_1 = e^{-0.8} = 0.4493$. Since $n = m$ there is no zero at infinity, so no $(z+1)$ factor pp. 14–15. DC: $\alpha \frac{1-0.8187}{1-0.4493} = 1$, so $\alpha = \frac{0.5507}{0.1813} = 3.038$.

(c) Divide by $7z$: $D = \dfrac{\frac{22}{7} - \frac{18}{7}z^{-1}}{1 - \frac{3}{7}z^{-1}}$. Direct programming pp. 104–107:

$$u(k) = \tfrac{3}{7}u(k-1) + \tfrac{22}{7}e(k) - \tfrac{18}{7}e(k-1).$$

$u(0) = \frac{22}{7} = \mathbf{3.143}$. $u(1) = \frac{3}{7}\cdot\frac{22}{7} + \frac{22}{7} - \frac{18}{7} = \frac{66}{49} + \frac{28}{49} = \frac{94}{49} = \mathbf{1.918}$. (It decays toward the DC gain 1, the lead "kick" you expect.)

> Common slips: writing $-\frac{3}{7}u(k-1)$ (the denominator sign must flip); using $2/T = 2$; adding a $(z+1)$ in (b) even though $n = m$.

Mark yourself (tick what your working shows):

- ☐ (a) substitution with $2/T = 20$
- ☐ (a) K_d, z_0, p_0 correct
- ☐ (b) mapped zero and pole, no $(z+1)$
- ☐ (b) $\alpha = 3.04$
- ☐ (c) difference equation with correct signs
- ☐ (c) $u(0)$, $u(1)$

B3. (10 marks) A unity-feedback loop has a digital controller $C(z)$ in front of

$$G_{ZAS}(z) = \frac{0.25(z+0.8)}{(z-1)(z-0.5)}.$$

The reference is a unit step.

(a) Show that $G_{cl}(z) = z^{-1}$ satisfies the design conditions, and find $C(z)$. [3]

(b) For the design in (a), find the control signal $u(k)$ for $k = 0, \ldots, 3$. Explain what happens to the continuous output between samples. [3]

(c) Design a ripple-free controller using $G_{cl}(z) = Kz^{-1}(1 + 0.8z^{-1})$. Find K and $C(z)$, and give $y(1)$ and $u(0)$. [4]

Final answers:

(b) $u(1) =$ [ ] × key -5.2 (b) $u(2) =$ [ ] × key 4.16

(c) $K =$ [ ] × key 0.5556 (c) $y(1) =$ [ ] × key 0.5556

(c) $u(0) =$ [ ] × key 2.222

Worked solution

(a) Check the four conditions pp. 51–56:

1. Relative degree: the plant's is 1, and z^{-1}'s is 1 ✓.
2. The plant zero -0.8 is inside the circle ✓.
3. The pole at $z = 1$ is marginal; $1 - z^{-1} = \frac{z-1}{z}$ contains it ✓.
4. $G_{cl}(1) = 1$ ✓.

$$C = \frac{1}{G_{ZAS}} \cdot \frac{z^{-1}}{1 - z^{-1}} = \frac{(z-1)(z-0.5)}{0.25(z+0.8)} \cdot \frac{1}{z-1} = \mathbf{\frac{4(z-0.5)}{z+0.8}}.$$

The plant already has an integrator, so C needs none.

(b) $U = \frac{G_{cl}}{G_{ZAS}} R = \frac{4(z-0.5)}{z+0.8}$, so $u = 4,\ -5.2,\ 4.16,\ -3.328, \ldots$, i.e. $u(k) = -5.2(-0.8)^{k-1}$ for $k \geq 1$. C cancels the plant zero at -0.8, so u keeps alternating and only decays slowly. The samples give $y(k) = 1$ for $k \geq 1$, but the continuous output **ripples between samples** (Example 5.8 pp. 66–67).

(c) Ripple-free means G_{cl} keeps *all* plant zeros, so that u settles in finitely many steps pp. 73–77. From $G_{cl}(1) = K(1 + 0.8) = 1$, $K = 1/1.8 = 5/9 = 0.5556$. Then

$$1 - G_{cl} = \frac{z^2 - \frac{5}{9}z - \frac{4}{9}}{z^2} = \frac{(z-1)(z+\frac{4}{9})}{z^2}, \qquad C = \frac{G_{cl}}{G_{ZAS}(1 - G_{cl})} = \mathbf{\frac{20}{9}} \cdot \mathbf{\frac{z - 0.5}{z + 0.4444}}.$$

$Y = G_{cl}R$ gives $y = 0,\ \frac{5}{9},\ 1,\ 1$, so $y(1) = 0.5556$. $U = \frac{G_{cl}}{G_{ZAS}} R = \frac{20}{9}(1 - 0.5z^{-1})$ gives $u = \frac{20}{9},\ -\frac{10}{9},\ 0,\ 0$, so $u(0) = 2.222$. u reaches 0 (the plant integrator holds $y = 1$), so there is no ripple. The cost is one extra sample to settle, and a gentler control signal.

> Common slips: $K = 1/0.8$ (the condition is $G_{cl}(1) = K(1 + 0.8) = 1$); saying (a) is ripple-free because $y(k) = 1$ at the samples; adding an integrator to C for a type-1 plant.

Mark yourself (tick what your working shows):

- ☐ (a) four conditions checked
- ☐ (a) $C(z) = 4(z - 0.5)/(z + 0.8)$
- ☐ (b) $u(k) = 4, -5.2, 4.16, -3.33$
- ☐ (b) ripple explained (cancelled zero at -0.8, u does not settle)
- ☐ (c) $K = 5/9$
- ☐ (c) $C(z)$ correct
- ☐ (c) $y(1) = 0.556$ and $u(0) = 2.22$, with $u \to 0$

Score 6/40. Section A (auto-marked): 6/12. Section B (your rubric ticks): 0/28. Green boxes match the key; red ones show the key beside them. Tick a rubric line only if your written working shows it. Then redo every wrong part on paper without looking.

Other papers: Mock C · Mock B (worked steps) · Primer